To learn these reverse trajectories, we need a lot of **intermediate noisy images.** These images, **which form the individual steps of the forward trajectories** (from clean data to noise), are generated by apply forward diffusion. The model when being trained with these intermediate noisy images, will try to learn the reverse trajectories.

The **resulting reverse trajectories resemble a high-dimensional vector field** that implicitly describe how to navigate from any point in the space towards the data manifold. The **data manifold is the conceptual space where all valid and realistic image samples exist**. Finding it is equivalent to revealing the underlying structure of training images.

**Directly calculating every possible trajectories from the intermediate noisy images to a realistic image** (the data manifold) **is practically impossible.** Moreover, we **don't even know what the target (realistic) image looks like beforehand.**

This makes a **direct, explicit computation intractable.** We can however, **train a model to learn and approximate the complex denoising trajectories.**

**Loss Function Derivation:**

The derivation begins with the **log-likelihood of the observed data points** (from the reverse process) **representing the real data**.

$$\frac{1}{N}\sum_{i}^{N} log_e p_\theta(x_{i,0})$$

For a data point, $x_{i,0}$, this likelihood can be expressed by considering the joint probability of $x_{i,0}$ and all the unobserved intermediate diffusion steps, $x_{i,1}, x_{i,2}, ..., x_{i,T}$ where $T$ is the total number of diffusion steps and $x_{i,T}$ is pure noise.

This process is equivalent to integrating out all the possible trajectories that could have led to the observed data point. This is also known as **marginalization**, as it effectively marginalizes out the unobserved variables $(x_{i,1}, x_{i,2}, ..., x_{i,T})$.

$$= \frac{1}{N}\sum_{i}^{N} log_e \int p_\theta(x_{i,0}, x_{i,1}, x_{i,2}, ..., x_{i,T})\, dx_{i,1} dx_{i,2} ... dx_{i,T}$$
$$= \frac{1}{N}\sum_{i}^{N} log_e \int p_\theta(x_{i,0:T})\, dx_{i,1:T}$$

Now, multiply and divide this expression by the **forward process distribution, *q*** which describes entire trajectory from $x_0$ to $x_t$ by chaining transition distributions.

$$= \frac{1}{N}\sum_{i}^{N} log_e \int p_\theta(x_{i,0:T})\, \frac{q(x_{i,1:T}|x_0)}{q(x_{i,1:T}|x_0)}\, dx_{i,1:T}$$

where, $q(x_{i,1:T}|x_0) = q(x_{i,1}|x_{i,0}).q(x_{i,2}|x_{i,1}).q(x_{i,3}|x_{i,2})...q(x_{i.T}|x_{i,T-1})$

**Expectation** of a function **when inputs are sampled from a probability distribution** is given by the following formula:

$$\mathbb{E}_{x \sim p(x)}[f(x)] = \int p(x) f(x) dx$$

where, $f(x) \longrightarrow$ value of the function and, $p(x) \longrightarrow$ how probable (likely) that value of $x$ is.

Reformulating our expression in terms of expection, we get

$$= \frac{1}{N}\sum_{i}^{N} \log_e \mathbb{E}_{x_{i,1:T} \sim q(x_{i,1:T}|x_0)} \left[ \frac{p_\theta(x_{i,0:T})}{q(x_{i,1:T}|x_0)} \right]$$

Since $logE[x] \geq E[log(x)]$ because of **Jensen's inequality**, and log is a concave function, Jensen's inequality ensures that

$$\geq \frac{1}{N}\sum_{i}^{N} \mathbb{E}_{x_{i,1:T} \sim q(x_{i,1:T}|x_0)} \left[ log_e \frac{p_\theta(x_{i,0:T})}{q(x_{i,1:T}|x_0)} \right]$$

Maximizing this objective (**ELBO**) leads to the maximization of the log likelihood.

---

**Notation Notes:**

- From this point forward, to avoid notational clutter, we'll simplify the terms by **focusing on a single observed data point,** $x$**,** and removing redundant notations.
- The emperical average (averaging over N data points) will be addressed at the end of the derivation.

**Preliminaries:**

1. $q(x_1, x_2|x_0)$ and $q(x_2, x_1|x_0)$ represent the same joint probability distribution.
2. The **diffusion process forms a Markov chain**. This means that each step depends only on the step before it. For example, $x_2$ depends only on $x_1$, $x_3$ depends only on $x_2$, and so on.
3. Using Condition probability, $p(a, b) = p(a).p(b|a)$

$$
\begin{aligned}
p(\underbrace{x_0}_{b}, \underbrace{x_1, x_2}_{a}) &= p(x_1, x_2) * p(x_0|x_1, x_2) \\
p(\underbrace{x_1}_{b}, \underbrace{x_2}_{a}) &= p(x_2) * p(x_1|x_2) \\
\implies p(x_0, x_1, x_2) &= p(x_0|x_1, x_2) * p(x_1|x_2) * p(x_2)
\end{aligned}
$$

$$
\boxed{p(x_0, x_1, \ldots, x_T) = p(x_0|x_{1:T})\ p(x_1|x_{2:T}) \ldots p(x_{T-1}|x_T)\ p(x_T)}
$$

4. Joint probability in terms of Conditional and Marginal probability:

$$
\begin{aligned}
q(\underline{x_3, x_2, x_1}, x_0) &= q(x_3, x_2, x_1|x_0)\, q(x_0) \\
q(\underline{x_3}, x_2, x_1, x_0) &= q(x_3|x_2, x_1, x_0)\, q(x_2|x_1, x_0)\, q(x_1|x_0)\, q(x_0) \\
\implies q(x_3, x_2, x_1|x_0) &= q(x_3|x_2, x_1, x_0)\, q(x_2|x_1, x_0)\, q(x_1|x_0)
\end{aligned}
$$

$$
\boxed{q(x_T, x_{T-1}, \ldots, x_1|x_0) = q(x_1|x_0)\, q(x_2|x_1, x_0)\ \ldots\ q(x_T|x_{T-1}, \ldots, x_1, x_0)}
$$

5. Standard 1D Gaussian Distribution in exponential form:

$$
f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)
$$

---

So, we have

$$
\mathbb{E}_q\left[log_e \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]
$$

We now expand this term:

$$
\begin{aligned}
&= \mathbb{E}_q\left[log_e \frac{p_\theta(x_0, x_1, ..., x_T)}{q(x_1, x_2, ..., x_T|x_0)}\right] \\
&= \mathbb{E}_q\left[log_e \frac{p_\theta(x_0, x_1, ..., x_T)}{q(x_T, x_{T-1}, ..., x_2, x_1|x_0)}\right] \\
&= \mathbb{E}_q\left[log_e \frac{p_\theta(x_0|x_1, ..., x_T)\ p_\theta(x_1|x_2, ..., x_T)...p_\theta(x_{T-1}|x_T)\ p_\theta(x_T)}{q(x_1|x_0)\ q(x_2|x_1, x_0)\ q(x_3|x_2, x_1, x_0)...q(x_T|x_{T-1}, ..., x_2, x_1, x_0)}\right] \\
&= \mathbb{E}_q\left[log_e \frac{p_\theta(x_0|x_1)\ p_\theta(x_1|x_2)...p_\theta(x_{T-1}|x_T)\ p_\theta(x_T)}{q(x_1|x_0)\ q(x_2|x_1)\ q(x_3|x_2)...q(x_T|x_{T-1})}\right]
\end{aligned}
$$

Rearrange it by applying $log(a.b) = log(a) + log(b)$

$$
\begin{aligned}
&= E_q\left[log_e\frac{p_\theta(x_0|x_1)}{q(x_1|x_0)} + log_e\frac{p_\theta(x_1|x_2)}{q(x_2|x_1)} + log_e\frac{p_\theta(x_{T-1}|x_T)}{q(x_T|x_{T-1})} + log_e p_\theta(x_T)\right] \\
&= E_q\left[log_e p_\theta(x_T) + \sum_{t=1}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}\right] \\
&= E_q\left[log_e p_\theta(x_T) + \sum_{t=2}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})} + log_e\frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}\right] \\
&= E_q\left[log_e p_\theta(x_T) + \sum_{t=2}^{T} log_e\frac{1}{q(x_t|x_{t-1})}\frac{p_\theta(x_{t-1}|x_t)}{1} + log_e\frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}\right] \\
&= E_q\left[log_e p_\theta(x_T) + \sum_{t=2}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)}\frac{q(x_{t-1}|x_0)}{q(x_t|x_0)} + log_e\frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}\right]
\end{aligned}
$$

Apply $log(a.b) = log(a) + log(b)$ to the second term,

$$
= E_q\left[log_e p_\theta(x_T) + \sum_{t=2}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)} + \sum_{t=2}^{T} log_e\frac{q(x_{t-1}|x_0)}{q(x_t|x_0)} + log_e\frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}\right]
$$

The second summation can further be simplified, as many terms in the numerator cancel with those in the denominator,

$$
= E_q\left[log_e p_\theta(x_T) + \sum_{t=2}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)} + log_e\frac{q(x_1|x_0)}{q(x_T|x_0)} + log_e\frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}\right]
$$

Apply $log(a.b) = log(a) + log(b)$ to simplify the expression,

$$
\begin{aligned}
&= E_q\left[log_e\frac{p_\theta(x_T)}{1}\frac{\cancel{q(x_1|x_0)}}{q(x_T|x_0)}\frac{p_\theta(x_0|x_1)}{\cancel{q(x_1|x_0)}} + \sum_{t=2}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)}\right] \\
&= E_q\left[log_e\frac{p_\theta(x_T)}{q(x_T|x_0)} + \sum_{t=2}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)} + log_e p_\theta(x_0|x_1)\right]
\end{aligned}
$$

Few keys adjustments are required to further simplify this expression. Firstly, the **distribution at the last step, $T$, can be simply approximated as a standard Gaussian prior**. This is because $x$ as $T$-th timestep is so corrupted by the iterative forward diffusion process that it is essentially just a random noise following standard Gaussian distribution.

$$
p_\theta(x_T) \Rightarrow p(x_T)
$$

Next, we need to **negate** the entire expression, as the negative log likelihood is commonly used as a loss functon for optimization,

$$
= E_q\left[-log_e\frac{p(x_T)}{q(x_T|x_0)} - \sum_{t=2}^{T} log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)} - log_e p_\theta(x_0|x_1)\right]
$$

Since both **summation and expectation are linear operators**, the expression can be written as,

$$
= E_q\left[-log_e\frac{p(x_T)}{q(x_T|x_0)}\right] + \sum_{t=2}^{T} E_q\left[-log_e\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)}\right] - E_q[log_e p_\theta(x_0|x_1)]
$$

The 1st and 2nd term can be converted into **Kullback-Leibler (KL) divergence** forms.

$$= D_{KL}\Big[q(x_T|x_0) \parallel p(x_T)\Big] + \sum_{t=2}^{T} D_{KL}\Big[q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t)\Big] - E_q[log_e p_\theta(x_0|x_1)]$$

For the entire dataset, this expression becomes,

$$= \frac{1}{N}\sum_{i}^{N} D_{KL}\Big[q(x_{i,T}|x_{i,0}) \parallel p(x_{i,T})\Big] + \sum_{t=2}^{T} D_{KL}\Big[q(x_{i,t-1}|x_{i,t}, x_{i,0}) \parallel p_\theta(x_{i,t-1}|x_{i,t})\Big] - E_{x_{i,1}\sim q(x_{i,1}|x_{i,0})}[log_e p_\theta(x_{i,0}|x_{i,1})]$$

**In generative modeling**, the **true data distribution,** $q(x_0)$, **is not available.** What we have is the finite dataset of N samples. We can approximate $q(x_0)$ using this data. This is known as **Monte Carlo estimation** or emperical estimation.

So in practice, the following expression is accepted:

$$E_{q(x_o)}[...] \approx \frac{1}{N}\sum_{i}^{N}[...]$$

Therefore, the expression becomes,

$$= E_{x_o\sim q(x_o)}\Bigg[D_{KL}\Big[q(x_T|x_0) \parallel p(x_T)\Big] + \sum_{t=2}^{T} D_{KL}\Big[q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t)\Big] - E_{x_1\sim q(x_1|x_0)}[log_e p_\theta(x_0|x_1)]\Bigg]$$

The third term is,

$$E_{x_o\sim q(x_o)}\Big[E_{x_1\sim q(x_1|x_o)}[-log_e p_\theta(x_0|x_1)]\Big] = E_{x_o\sim q(x_o),\ x_1\sim q(x_1|x_0)}[-log_e p_\theta(x_0|x_1)]$$

Since $x_0$ and $x_1$ are **very similar**, especially with small noise additions, we can **collapse the nested expectation into a single outer expectation** over $x_0 \sim q(x_0)$. This allows for the term to be sampled just once. This leads to the simplified final DDPM ELBO objective.

$$\boxed{= E_{x_o\sim q(x_o)}\Bigg[D_{KL}\Big[q(x_T|x_0) \parallel p(x_T)\Big] + \sum_{t=2}^{T} D_{KL}\Big[q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t)\Big] - log_e p_\theta(x_0|x_1)\Bigg]}$$

This ELBO objective function consists of two KL terms and one reconstruction term. The authors of the original DDPM paper simplified this expression even further. The authors found that since $x_0$ and $x_1$ are very similar, the reconstruction term contributes little to the overall training signal. They dropped it during training. In addition, they also discarded the first KL term because it does not involve any of the model parameters; thus, minimizing it has no effect on the optimization of the model.

Therefore, the expression becomes,

$$= E_{x_o\sim q(x_o)}\Bigg[\cancel{D_{KL}\Big[q(x_T|x_0) \parallel p(x_T)\Big]} + \sum_{t=2}^{T} D_{KL}\Big[q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t)\Big] - \cancel{log_e p_\theta(x_0|x_1)}\Bigg]$$

$$\boxed{= E_{x_o\sim q(x_o)}\Bigg[\sum_{t=2}^{T} D_{KL}\Big[q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t)\Big]\Bigg]}$$

**For each step *t*,** it **compares the true posterior** q **conditioned on** $x_0$ **with the learned reverse distribution** $p_\theta$. This is simply **requiring approximate denoising transition distribution to be very close to the ground truth denoising transition distribution conditioned on** $x_0$.

---

Now we will compute $q(x_{t-1}|x_t,x_0)$

Using Conditional Probability, we get,

$$
\begin{aligned}
q(x_{t-1}|x_t,x_0) &= \frac{q(x_0,x_{t-1},x_t)}{q(x_0,x_t)} \\
\implies q(x_{t-1}|x_t,x_0) &= \frac{q(x_0,x_{t-1})\ q(x_t|x_0,x_{t-1})}{q(x_0,x_t)} \\
\implies q(x_{t-1}|x_t,x_0) &= \frac{\cancel{q(x_0)}\ q(x_{t-1}|x_0)\ q(x_t|x_0,x_{t-1})}{\cancel{q(x_0)}\ q(x_t|x_0)} \\
\implies q(x_{t-1}|x_t,x_0) &= \frac{q(x_t|x_{t-1},x_0)\ q(x_{t-1}|x_0)}{q(x_t|x_0)} \\
\implies q(x_{t-1}|x_t,x_0) &= \frac{\mathcal{N}(x_t;\sqrt{\alpha}_t x_{t-1},(1-\alpha_t)I)\ \mathcal{N}(x_{t-1};\sqrt{\overline{\alpha}_{t-1}}x_0,(1-\overline{\alpha}_{t-1})I)}{\mathcal{N}(x_t;\sqrt{\overline{\alpha}_t}x_0,(1-\overline{\alpha}_t)I)}
\end{aligned}
$$

We will now write these Gaussians in exponential form. We will **ignore** the **normalization (scaling) terms** because they are **constant** (DDPM authors used a fixed variance) and **do not depend on the model parameters** $\theta$. Therefore, they **do not affect the optimization**.

$$
\begin{aligned}
&= exp\frac{-1}{2}\left[\frac{(x_t-\sqrt{\alpha}_t x_{t-1})^2}{1-\alpha_t}+\frac{(x_{t-1}-\sqrt{\overline{\alpha}_{t-1}}x_0)^2}{1-\overline{\alpha}_{t-1}}-\frac{(x_t-\sqrt{\overline{\alpha}_t}x_0)^2}{1-\overline{\alpha}_t}\right] \\
&= exp\frac{-1}{2}\left[\frac{x_t^2-2\sqrt{\alpha}_t x_{t-1}x_t+\alpha_t x_{t-1}^2}{1-\alpha_t}+\frac{x_{t-1}^2-2\sqrt{\overline{\alpha}_{t-1}}x_0x_{t-1}+\overline{\alpha}_{t-1}x_0^2}{1-\overline{\alpha}_{t-1}}-\frac{x_t^2-2\sqrt{\overline{\alpha}_t}x_0x_t+\overline{\alpha}_t x_0^2}{1-\overline{\alpha}_t}\right] \\
&= exp\frac{-1}{2}\left[\left(\frac{x_t^2}{1-\alpha_t}-\frac{2\sqrt{\alpha}_t x_{t-1}x_t}{1-\alpha_t}+\frac{\alpha_t x_{t-1}^2}{1-\alpha_t}\right)+\left(\frac{x_{t-1}^2}{1-\overline{\alpha}_{t-1}}-\frac{2\sqrt{\overline{\alpha}_{t-1}}x_0x_{t-1}}{1-\overline{\alpha}_{t-1}}+\frac{\overline{\alpha}_{t-1}x_0^2}{1-\overline{\alpha}_{t-1}}\right)-\left(\frac{x_t^2}{1-\overline{\alpha}_t}-\frac{2\sqrt{\overline{\alpha}_t}x_0x_t}{1-\overline{\alpha}_t}+\frac{\overline{\alpha}_t x_0^2}{1-\overline{\alpha}_t}\right)\right] \\
&= exp\frac{-1}{2}\left[x_{t-1}^2\left(\frac{\alpha_t}{1-\alpha_t}+\frac{1}{1-\overline{\alpha}_{t-1}}\right)-2x_{t-1}\left(\frac{\sqrt{\alpha}_t x_t}{1-\alpha_t}+\frac{\sqrt{\overline{\alpha}_{t-1}}x_0}{1-\overline{\alpha}_{t-1}}\right)+\left(\frac{x_t^2}{1-\alpha_t}+\frac{\overline{\alpha}_{t-1}x_0^2}{1-\overline{\alpha}_{t-1}}-\frac{x_t^2}{1-\overline{\alpha}_t}+\frac{2\sqrt{\overline{\alpha}_t}x_0x_t}{1-\overline{\alpha}_t}-\frac{\overline{\alpha}_t x_0^2}{1-\overline{\alpha}_t}\right)\right] \\
&= exp\frac{-1}{2}\left[x_{t-1}^2\left(\frac{\alpha_t(1-\overline{\alpha}_{t-1})+(1-\alpha_t)}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\right)-2x_{t-1}\left(\frac{\sqrt{\alpha}_t x_t(1-\overline{\alpha}_{t-1})\ +\ (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\right)+(...)\right] \\
&= exp\frac{-1}{2}\left[x_{t-1}^2\frac{1-\overline{\alpha}_t}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}-2x_{t-1}\frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha}_t x_t+(1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}+(...)\right] \\
&= exp\frac{-1}{2}\left[\frac{1-\overline{\alpha}_t}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\left\{x_{t-1}^2-2x_{t-1}\frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha}_t x_t+(1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{1-\overline{\alpha}_t}+(...)\right\}\right]
\end{aligned}
$$

Let us now calcuate the last term: (...),

$$
\begin{aligned}
&= \frac{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t}.\left(\frac{x_t^2}{1-\alpha_t}+\frac{\overline{\alpha}_{t-1}x_0^2}{1-\overline{\alpha}_{t-1}}-\frac{x_t^2}{1-\overline{\alpha}_t}+\frac{2\sqrt{\overline{\alpha}_t}x_0x_t}{1-\overline{\alpha}_t}-\frac{\overline{\alpha}_t x_0^2}{1-\overline{\alpha}_t}\right) \\
&= x_t^2\frac{(1-\overline{\alpha}_{t-1})(\alpha_t-\overline{\alpha}_t)}{(1-\overline{\alpha}_t)(1-\overline{\alpha}_t)}\ +\ x_0^2\frac{(1-\alpha_t)(\overline{\alpha}_{t-1}-\overline{\alpha}_t)}{(1-\overline{\alpha}_t)(1-\overline{\alpha}_t)}\ +\ \frac{2\sqrt{\overline{\alpha}_t}x_0x_t(1-\alpha_t)(1-\overline{\alpha}_{t-1})}{(1-\overline{\alpha}_t)(1-\overline{\alpha}_t)} \\
&= x_t^2\frac{(1-\overline{\alpha}_{t-1})^2\ \alpha_t}{(1-\overline{\alpha}_t)^2}\ +\ x_0^2\frac{(1-\alpha_t)^2\ \overline{\alpha}_{t-1}}{(1-\overline{\alpha}_t)^2}\ +\ \frac{2\sqrt{\alpha_t}\sqrt{\overline{\alpha}_{t-1}}x_0x_t(1-\alpha_t)(1-\overline{\alpha}_{t-1})}{(1-\overline{\alpha}_t)^2} \\
&= \left[\frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha_t}x_t\ +\ (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{(1-\overline{\alpha}_t)}\right]^2
\end{aligned}
$$

Putting it back into our expression, we get,

$$= exp\frac{-1}{2}\left[\frac{1-\overline{\alpha}_t}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\left\{x_{t-1}^2 - 2x_{t-1}\frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha}_t x_t + (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{1-\overline{\alpha}_t} + \left[\frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha_t}x_t \ + \ (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{(1-\overline{\alpha}_t)}\right]^2\right\}\right]$$

$$\boxed{= exp\frac{-1}{2}\left[\frac{1-\overline{\alpha}_t}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\left\{x_{t-1} - \frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha}_t x_t + (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{1-\overline{\alpha}_t}\right\}^2\right]}$$

---

**Note:**

A **scaler Gaussian PDF** can be written in either of these equivalent ways:

- **Standard (Mean-Variance) Form**

$$\mathcal{N}(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma}exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$$

- **Quadratic (Precision) Form**

$$p(x) \propto exp\Big(-\frac{1}{2}ax^2 + bx + c\Big), a > 0$$

---

Grouping the quadratic and linear terms, we get:

$$-\frac{1}{2}ax^2 + bx + c \ = \ -\frac{1}{2}\Big(ax^2 - 2bx - 2c\Big) \ = -\frac{1}{2}a\Big(x - \frac{b}{a}\Big)^2 + \frac{b^2}{2a} + c$$

Plugging it into the exponential we get:

$$p(x) \propto exp\Bigg(-\frac{1}{2}a\Big(x-\frac{b}{a}\Big)^2 + \frac{b^2}{2a} + c\Bigg)$$
$$\implies p(x) \propto exp\Bigg(-\frac{1}{2}a\Big(x-\frac{b}{a}\Big)^2\Bigg).exp\Bigg(\frac{b^2}{2a} + c\Bigg)$$

Since the constant term can be absorbed into normalization constant, the resulting expression becomes:

$$p(x) \propto exp\Bigg(-\frac{1}{2}a\Big(x-\frac{b}{a}\Big)^2\Bigg)$$

Comparing it with the standard Gaussian form, we get:

$$\sigma^2 = \frac{1}{a}, \quad \mu = \frac{b}{a}$$

Our expression was:

$$exp\left[\frac{-1}{2}\underbrace{\frac{1-\overline{\alpha}_t}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}}_{a}\left\{x_{t-1} - \underbrace{\frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha}_t x_t + (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{1-\overline{\alpha}_t}}_{\frac{b}{a}}\right\}^2\right]$$

We can rewrite our expression as,

$$\boxed{\implies q(x_{t-1}|x_t,x_0) = \mathcal{N}\Bigg(x_{t-1};\underbrace{\frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha_t}x_t + (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{1-\overline{\alpha}_t}}_{\mu_q(x_t,x_0)}, \underbrace{\frac{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t}}_{\Sigma_q(t)}I\Bigg)}$$

This expression contains $x_0$ which won't be availabe to us during generation. Since we know that the **reverse of a Gaussian process is also a Gaussian**, we can approximate $p_\theta(x_{t-1}|x_t)$ also as a Gaussian. All we need to do is to **learn the mean and variance**.

$$p_\theta(x_{t-1}|x_t) \;\longrightarrow\; \mathcal{N}\Big(x_{t-1}; \mu_\theta(x_t), \Sigma_\theta(t)\Big)$$

The DDPM authors **fixed the variance exactly same as the ground truth denoising step variance.**

$$\Longrightarrow \boxed{p_\theta(x_{t-1}, x_t) \;\longrightarrow\; \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t), \frac{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t} I\right)}$$

---

Let $p(x)$ and $q(x)$ be two Normal distributions **with same variance**. The KL-divergence between them is defined as:

$$D_{KL}(p\|q) = E_p\left[log\frac{p(x)}{q(x)}\right]$$

In standard Multivariate Gaussian form (where $x \in R^d$),

$$p(x) = \frac{1}{(2\pi)^{d/2}\,\left|\Sigma_p\right|^{1/2}} exp\left[-\frac{1}{2}(x-\mu_p)^T\Sigma_p^{-1}(x-\mu_p)\right]$$

$$q(x) = \frac{1}{(2\pi)^{d/2}\,\left|\Sigma_q\right|^{1/2}} exp\left[-\frac{1}{2}(x-\mu_q)^T\Sigma_q^{-1}(x-\mu_q)\right]$$

For isotropic covariance (where all dimensions are independent and have same variance), $\Sigma = \sigma^2 I$, $\left|\Sigma\right| = (\sigma^2)^d$

Taking log on both sides,

$$ln[p(x)] = -\frac{d}{2}log[\sqrt{2\pi\sigma^2}\,] - \frac{1}{2\sigma^2}\left\|x-\mu_p\right\|^2$$

$$ln[q(x)] = -\frac{d}{2}log[\sqrt{2\pi\sigma^2}\,] - \frac{1}{2\sigma^2}\left\|x-\mu_q\right\|^2$$

Subtracting $ln[q(x)]$ from $ln[p(x)]$, we get

$$ln\left[\frac{p(x)}{q(x)}\right] = -\frac{1}{2\sigma^2}\left(\left\|x-\mu_p\right\|^2 - \left\|x-\mu_q\right\|^2\right)$$

Expand the norm terms,

$$\left\|x-\mu_p\right\|^2 - \left\|x-\mu_q\right\|^2 = (x^Tx - 2x^T\mu_p + \mu_p^T\mu_p) \;-\; (x^Tx - 2x^T\mu_q + \mu_q^T\mu_q) = -2x^T(\mu_p-\mu_q) + \left\|\mu_p\right\|^2 - \left\|\mu_q\right\|^2$$

Plugging the above norm equation back into the KL expression, we get,

$$ln\left[\frac{p(x)}{q(x)}\right] = -\frac{1}{2\sigma^2}\left(-2x^T(\mu_p-\mu_q) + \left\|\mu_p\right\|^2 - \left\|\mu_q\right\|^2\right)$$

Taking expectation on both sides, we get:

$$D_{KL}(p\|q) = -\frac{1}{2\sigma^2}E_p\left[-2x^T(\mu_p-\mu_q) + \left(\left\|\mu_p\right\|^2 - \left\|\mu_q\right\|^2\right)\right]$$

$$\Longrightarrow D_{KL}(p\|q) = \frac{1}{\sigma^2}E_p\left[x^T(\mu_p-\mu_q)\right] - \frac{1}{2\sigma^2}\left(\left\|\mu_p\right\|^2 - \left\|\mu_q\right\|^2\right)$$

Since $x \sim N(\mu_p, \sigma^2 I) \Rightarrow E_p[x] = \mu_p$. Therefore $E_p[x^T(\mu_p - \mu_q)] = \mu_p^T(\mu_p - \mu_q)$

Substituting it back we get,

$$D_{KL}(p\|q) = \frac{1}{\sigma^2}\mu_p^T(\mu_p - \mu_q) - \frac{1}{2\sigma^2}\left(\|\mu_p\|^2 - \|\mu_q\|^2\right)$$

After simplification, the KL expression becomes

$$D_{KL}(p\|q) = \frac{1}{2\sigma^2}\|\mu_p - \mu_q\|^2$$

$$\implies \boxed{D_{KL}(p\|q) = \frac{1-\overline{\alpha}_t}{2(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\|\mu_p - \mu_q\|^2}$$

We can adapt this general equation to our case,

$$\implies \boxed{D_{KL}\Big(\mathcal{N}(x_{t-1}; \mu_\theta(x_t), \Sigma_q(t)) \parallel \mathcal{N}(x_{t-1}; \mu_q(x_t, x_0), \Sigma_q(t))\Big) = \frac{1-\overline{\alpha}_t}{2(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\|\mu_\theta(x_t) - \mu_q(x_t, x_0)\|^2}$$

---

Our ground truth denoising step has mean in the following form:

$$\mu_q(x_t, x_0) = \frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha_t}x_t + (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_0}{1-\overline{\alpha}_t}$$

We can rewrite the mean for our approximate denoising distribution also in this form. However, since during reverse sampling, $x_0$ is unknown, we'll have to approximate it.

$$\mu_\theta(x_t) = \frac{(1-\overline{\alpha}_{t-1})\sqrt{\alpha_t}x_t + (1-\alpha_t)\sqrt{\overline{\alpha}_{t-1}}x_\theta}{1-\overline{\alpha}_t}$$

Here, $x_\theta$ is the **approximation of the original image base on what $x_t$ looks like**.

The closed form of forward process is described as:

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$\implies x_0 = \frac{1}{\sqrt{\overline{\alpha}_t}}(x_t - \sqrt{1-\overline{\alpha}_t}\epsilon)$$

Since our model will be **trained to predict the noise that was added to a clean image to get image at timestep** $t$, we can write $x_\theta$ in terms of predicted noise.

$$x_\theta = \frac{1}{\sqrt{\overline{\alpha}_t}}(x_t - \sqrt{1-\overline{\alpha}_t}\epsilon_\theta(x_t, t))$$

Plugging $x_0$ into $\mu_q(x_t, x_0)$ and $x_\theta$ into $\mu_\theta(x_t)$, we get

$$\boxed{\mu_q(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon\right)} \; and \; \boxed{\mu_\theta(x_t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(x_t, t)\right)}$$

Using this estimated (model) mean would **ensure that the reverse process matches the analytical posterior structure while depending only on the observable variables**.

Now we calculate, $\mu_\theta(x_t) - \mu_q(x_t, x_0)$

$$
\begin{aligned}
&= \frac{1}{\sqrt{\alpha}_t}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(x_t,t)\right) \;-\; \frac{1}{\sqrt{\alpha}_t}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon\right) \\
&= \frac{1}{\sqrt{\alpha}_t}\left(\cancel{x_t} - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(x_t,t) \;-\; \cancel{x_t} \;+\; \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon\right) \\
\implies &\boxed{\mu_\theta(x_t) - \mu_q(x_t,x_0) = \frac{1-\alpha_t}{\sqrt{\alpha_t}\sqrt{1-\overline{\alpha}_t}}\Big(\epsilon \;-\; \epsilon_\theta(x_t,t)\Big)}
\end{aligned}
$$

Earlier, we got the following expression:

$$
D_{KL}\Big(\mathcal{N}(x_{t-1};\mu_\theta(x_t),\Sigma_q(t)) \parallel \mathcal{N}(x_{t-1};\mu_q(x_t,x_0),\Sigma_q(t))\Big) = \frac{1-\overline{\alpha}_t}{2(1-\alpha_t)(1-\overline{\alpha}_{t-1})}\Big\|\mu_\theta(x_t) - \mu_q(x_t,x_0)\Big\|^2
$$

$$
\implies \boxed{D_{KL}\Big(\mathcal{N}(x_{t-1};\mu_\theta(x_t),\Sigma_q(t)) \parallel \mathcal{N}(x_{t-1};\mu_q(x_t,x_0),\Sigma_q(t))\Big) = \frac{1}{2\sigma_q^2}\frac{(1-\alpha_t)^2}{\alpha_t(1-\overline{\alpha_t})}\Big\|\epsilon_\theta(x_t,t) \;-\; \epsilon\Big\|^2}
$$

The **entire front factor depends only on t, so during training, minimzing KL is equivalent to minimizing,**

$$
\boxed{\Big\|\epsilon_\theta(x_t,t) \;-\; \epsilon\Big\|^2}
$$

## References


- J. Park, *From Noise to Clarity: The Math Behind Denoising Diffusion Models* Medium article
- DDPM derivation tutorial YouTube video